

【総合問題演習 特別編】

次の英文を読んで、以下の設問に答えよ。

What is the connection between mind and brain? This is not (1a) a scientific issue; it is also a philosophical question, and an ancient one as well. Over the next few decades, (2)as scientists reveal the mechanism of the brain in greater detail, the question of the connection between our brains and our minds will become a more urgent matter for further discussion.

The brain is a physical system. It (3a) about 100 billion interconnected neurons — about as many neurons as there are stars in the Milky Way. (4)It is not the number of cells that is important here, but the connections between them. Each neuron may receive signals from thousands of others, and may, in turn, send signals out to thousands more. The neurons seem to be arranged (1b): those that receive signals from the senses process them and (3b) them on to higher systems of neurons. In the end, by mechanisms we still do not fully understand, these signals are converted by neurons in different parts of the brain into the final signals that produce images, smells or sounds. Thus the brain works (1c) by passing information from neuron to neuron. (5)The goal of brain research would be to attain the detailed knowledge of how neurons function, or in other words, which neurons are firing in any circumstance.

The fundamental question then becomes what the mind is. Formal definitions usually mention something like "the sum of mental activities," but (6)that does not tell us much. On the other hand, we all have had the experience of mind. Close your eyes and think of an episode from your childhood. You (1d) can have a fairly detailed visual image of some setting, maybe even some sounds and smells. You have these images "in mind," but where, exactly, are they? They (1e) do not correspond to any input from the senses into your brain right now, even though they must involve the firing of neurons, somewhere. In the mid-twentieth century neurosurgeons operating on the brains of conscious patients discovered that they could produce these sorts of images by applying electrical current to specific parts of the brain. Obviously, there is some sort of connection between the activity of neurons and our ability to create images, which we (3c) with the concept of "mind."

But what can that connection be? Who is the "I" that says "I remember," and where is he or she (3d)? One (7) made by René Descartes, "I think, therefore I am." In his view, the mind is something that controls the physical brain. However, such a view that the mind is beyond or separated from the brain is still controversial.

(8)If this view of the mind as something that is beyond or separated from the brain is not correct, then the firing of neurons is all there is. This, however, does not work, either. (3e) that at some point in the future a neuroscientist could say, "When you see the color blue, this particular set of neurons will fire in this particular order." Suppose that every time you saw blue, those particular neurons fired, and

that they never fired in the same way when you saw anything else. Clearly, you would have established a correlation between the experience of seeing blue and a particular process in the brain.

But, even if you could establish the correlation, you would not have explained the experience itself!

(9) You are not, after all, aware of neurons firing — you are aware of the color blue, and the most brilliant neurological research in the world cannot bridge that gap.

To understand the experience itself, (10) — psychology! A lot of work is being done to understand human perception and cognition. There are two groups of people trying to bridge the gap between mind and brain. On the one side, working up from the smallest scale, are the neuroscientists. On the other, working down from the largest scale, are the psychologists. (11) Whether the two will ever come together is very much an open question.

設 問

(1) 本文中の空欄(1a), (1b), (1c), (1d), (1e)に当てはまる語を以下の選択肢から 1 つずつ選び, 1a—○○のように答えよ。ただし, 同じ語を複数回選んではならない。

probably / hierarchically / purely / obviously / basically

(2) 以下の小問(a), (b)に答えよ。

(a) 下線部(2)冒頭の as の品詞を答えたいえで, そのように判断した理由を 1 行以内で説明せよ。

(b) 下線部(2)を和訳せよ。

(3) 本文中の空欄(3a), (3b), (3c), (3d), (3e)に当てはまる語を以下の選択肢から 1 つずつ選び, 必要に応じて語形を変化させて, 3a—○○のように答えよ。ただし, 同じ語を複数回選んではならない。また, 文頭にくる語も小文字で記してあるので注意せよ。

suppose / associate / contain / locate / pass

(4) 下線部(4)を和訳せよ。

(5) 下線部(5)を和訳せよ。

(6) 下線部(6)とはどういうことか, 2 行以内で説明せよ。

(7) 下に与えられた単語を正しい順に並べ替え, 空欄(7)を埋めるのに最も適切な表現を完成させよ。

the / this / of / remark / way / consider / to / at / is / question / looking

(8) 下線部(8)を和訳せよ。

(9) 下線部(9)を和訳せよ。ただし, that gap の内容を明示すること。

(10) 下に与えられた単語を正しい順に並べ替え, 空欄(10)を埋めるのに最も適切な表現を完成させよ。

have / in / you / science / a / different / of / to / start / branch / wholly

(11) 下線部(11)を和訳せよ。ただし, the two の内容を明示すること。

2024/2025

解答は、設問・小問ごとに改行し、冒頭に設問番号((1), (2), ...)を付して記入せよ。

[illegible]

[illegible]